

Causal Inference for Panel Data

An applied approach

Day 1

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9th and 10th December 2025

University of Southampton

Course introduction

Who I am

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 - Some tips to overcome them

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- The main aim of this course is to **share some of my experience** with you:
 - Main issues and problems often encountered in applied work
 - Some tips to overcome them
- We will treat all the topics of this course mainly from an **applied perspective**

Overview of the course

This course is organised over 2 days:

Day 1

- December 9th from 9:45 to 12:15(ish)
- Building 58 Room 2097 and online
- **Topic 1:** Definition and features of panel data
- **Topic 2:** Difference-in-differences: introduction and review of the up-to-date literature
- **Topic 3:** Longitudinal datasets available on the UK (optional)

Overview of the course

Day 2

- December 10th from 10 to 13
- Building 58 Room 2097 and online
- Lab session using **Stata**
- **Part 1:** Problem set 1 - guided
- **Part 2:** Problem set 2 - guided
- **Part 3:** Problem set 3 - autonomous

Today's schedule

Day 1 - Schedule

- 9:45 → Introduction to this course
- 10:00 → Panel data - why so useful for causal inference
- 10:45 → DiD - definition and assumptions
- 11:15 → Break
- 11:30 → DiD in practice - overview of the new literature
- 11:45 → UK longitudinal datasets (optional)

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Otherwise it becomes a boring monologue :)

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1. Introduction
2. Panel data - definition
3. Panel data in practice
4. Difference-in-differences
5. Instructions for tomorrow
6. UK longitudinal data

The issue of causality

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- How **public policies** affect poverty/unemployment etc?
- Does **higher education** lead to better jobs?
- Will **attend this Winter School** make me a more successful researcher?

The issue of causality

- Applied research → Use of **statistical methods and tools** to uncover relationships between variables

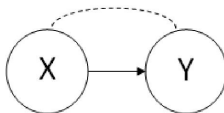
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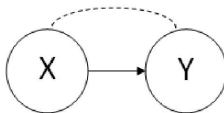
Causation



Changes in x cause changes in y.

The issue of causality

Causation



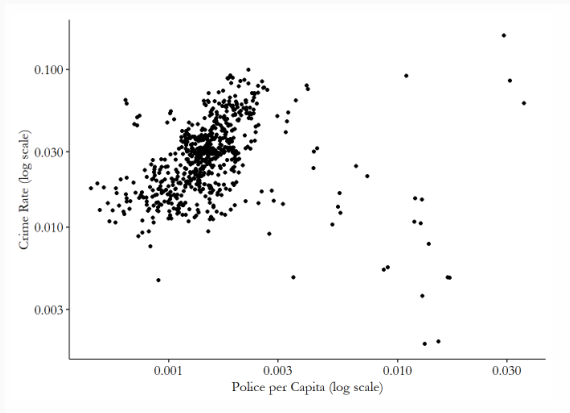
Changes in x cause changes in y.

E.g. Sending additional police in the streets will decrease crime

The issue of causality

However, reality is much more **complex**:

Figure 1: Police Presence and Crime Rate by County, North Carolina 1981-1987



Source: theeffectbook.net

Panel data - definition

Experimental data (RCT)

- Ideal scenario
- Designed to evaluate a specific policy/treatment and investigate causal effect

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- Ideal scenario
- Designed to evaluate a specific policy/treatment and investigate causal effect
- Often not feasible and difficult to set up
- Can present issues of external validity

Observational data

- Data obtained outside an experimental setting
- Survey data, administrative data, historical records
- Combine different econometrics techniques with clear identification strategy
- Mostly used in applied work

Quasi-Experimental Approaches

- Use observational data and find "natural experiments"
- Need to motivate identification strategy and find a clear exogenous variation
- Useful econometrics tool: Difference-in-differences, RDD, IV

Cross-sectional data

- Multiple units, only one time

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Notation

Y_i and X_i where $i = 1, \dots, N$

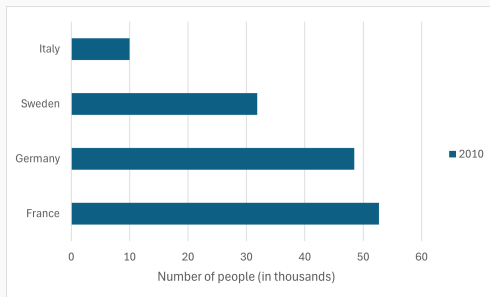
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Figure 2: Total number of asylum applicants in 2010 in selected EU countries



Source: Asylum applicants by type, citizenship, age and sex - annual aggregated data, Eurostat

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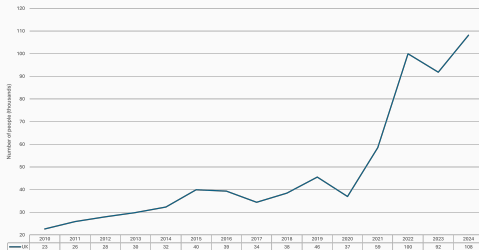
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Figure 3: Trends in total number of people claiming asylum in the UK



Source: Asy_D01 - Asylum applications raised, Home Office

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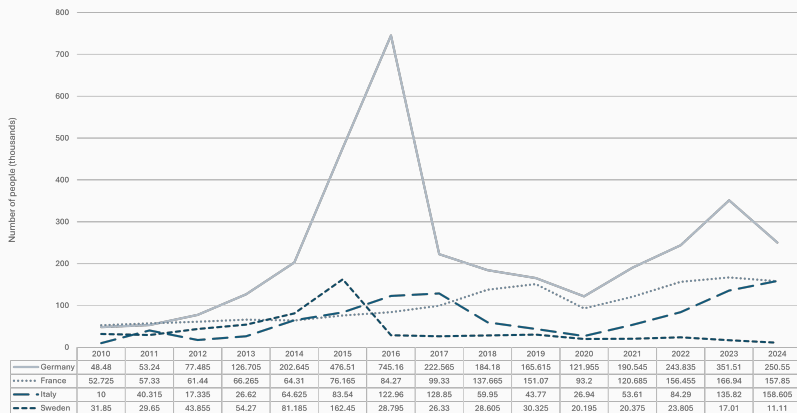
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Y_{it} or X_{it} for $i = 1, \dots, N$ and $t = 1, \dots, T$

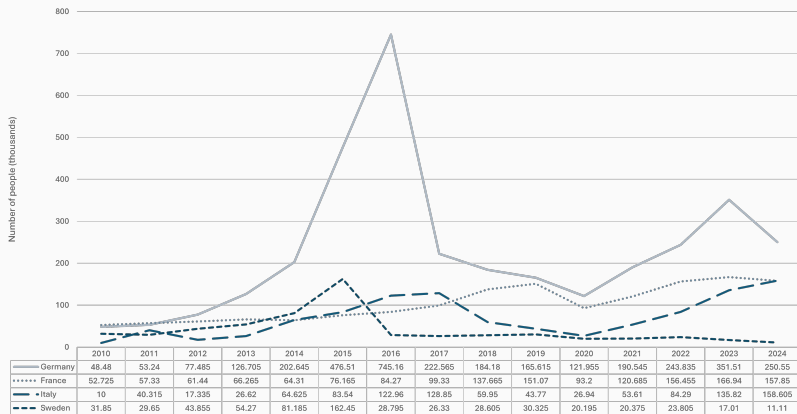
Panel data - example

Figure 4: Trends in total number of asylum applicants in selected EU countries



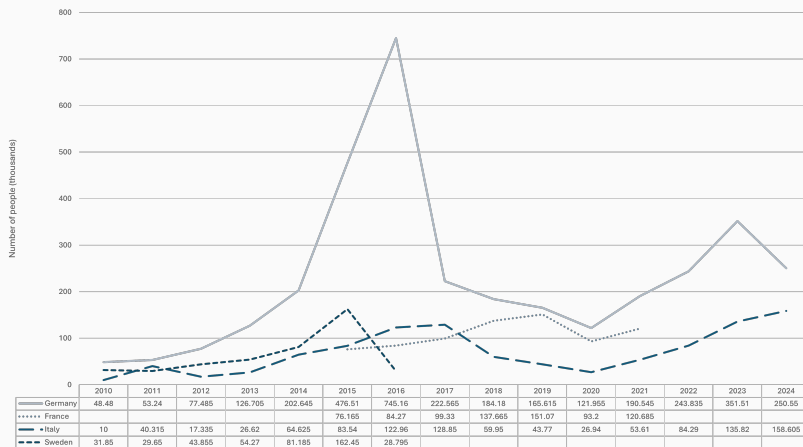
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Figure 5: Trends in total number of asylum applicants in selected EU countries



Source: Asylum applicants by type, citizenship, age and sex - annual aggregated data, Eurostat

Figure 6: Trends in total number of asylum applicants in selected EU countries



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Balanced vs unbalanced

A panel is **balanced** if we have the same time periods, $t = 1, \dots, T$, for each cross section observation. A panel is **unbalanced** if the time dimension is not the same to each individual.

- Issue of **sample selection and attrition**
- **Attrition**: when individuals drop out of the sample

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Micro vs Macro

- **Micro**: More i than t
- **Macro**: More t than i or roughly the same

Panel data in practice

Panel data: main advantage

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$$Y_{it} = \alpha_i + \beta_1 X_{it} + \epsilon_{it} \quad (1)$$

OLS is only consistent if X_{it} is uncorrelated with the error ϵ_{it}

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$$\epsilon_{it} = \mu_i + \gamma_{it} \quad (3)$$

- μ_i unit-specific invariant characteristics correlated with our treatment
- γ_{it} idiosyncratic part of the error that is NOT correlated with the treatment

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Fixed effects will only eliminate μ_i , the part of the error that is constant over time

Defining your equation in practice

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- Your parameter of interest is β_1

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- Include units specific dummies
- Use one of the software commands (recommended)

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- Two-way fixed effects: a method that includes fixed effects for both the cross-sectional and time unit
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- E.g. including a dummy for the year 2020 will control for Covid
- With `xtreg` need to include time dummies (e.g. `i.year`)

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TWFE equation

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- Y_{it} is the outcome variable
- α_i is the constant
- X_{it} is your variable of interest
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```
xtreg depvar indepvar, fe vce(cluster id)
```

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- More **computationally efficient**
- Stata command → ***reghdfe depvar indvar, absorb() vce()***

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- Virtually more observations
- Multiple points in time → you can use leads and lags
- You can consider using random effect to estimate causal relationships if you have multi-level data → more on this

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- With non-linear model → **incidental parameter problem**

Practical example

I want to estimate the causal effect of the number of published papers on the probability of getting a job in academia

$$Y_{it} = \alpha_i + \beta_1 X_{it} + \epsilon_{it} \quad (6)$$

- Which variables would you include as control variables?
- What could be an example of time-invariant heterogeneity that can be difficult to observe in real data?
- And time variant?
- Are there any issues of reverse causality in this relationship?

Difference-in-differences

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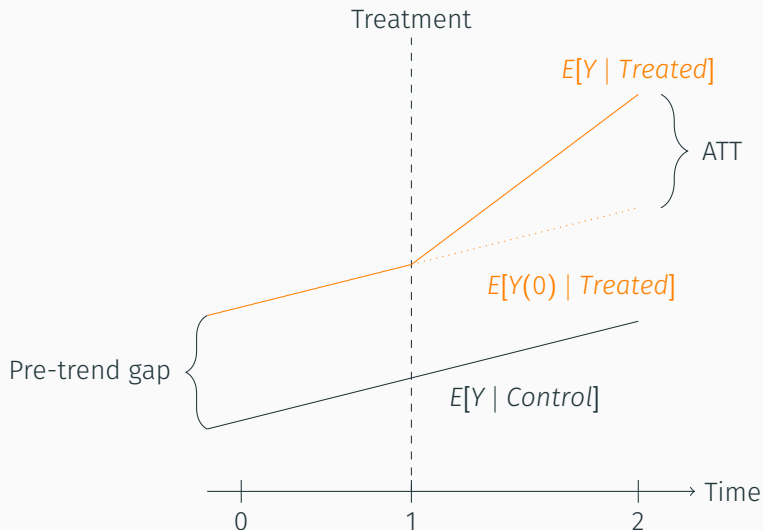
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- Group of units that were not exposed to the policy/intervention → Control group
- Observe the outcomes of treatment and control group before and after the policy/intervention took place
- Need panel data

DiD introduction by Angrist

DiD what are we estimating?



Source: Adapted from **Mixtape - Advanced DiD**

$$Y_{it} = \alpha + \beta_1 Treated_i + \beta_2 Post_t + \beta_3 (Treated_i \times Post_t) + \epsilon_{it} \quad (7)$$

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Standard DiD in practice

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- β_3 = DiD estimate $\rightarrow$ the effect of the treatment on the treated group after the treatment

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- $Interaction = Treated_i \times Post_t$
- β_3 = DiD estimate $\rightarrow$ the effect of the treatment on the treated group after the treatment
- The DiD is the gap in the slopes

$$Y_{it} = \beta_3(Treated_i \times Post_t) + \mu_i + \delta_t + \epsilon_{it} \quad (8)$$

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- δ_t = time fixed effects $\rightarrow$ controls for shocks common to all units at time t

$$Y_{it} = \beta_3(Treated_i \times Post_t) + \mu_i + \delta_t + \epsilon_{it} \quad (8)$$

- μ_i = unit fixed effects $\rightarrow$ controls for all time-invariant differences across units
- δ_t = time fixed effects $\rightarrow$ controls for shocks common to all units at time t
- β_3 = Coefficient of the interaction $\rightarrow$ DiD effect

- No anticipation

Elements of the standard DiD model

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- All units treated at the same time

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- Absence of pre-trends → key identifying assumption

Elements of the standard DiD model

- No anticipation
- All units treated at the same time
- Absence of pre-trends → key identifying assumption
 - Visual inspection of pre-treatment trends
 - Event-study models
 - Placebo tests

Common ways to assess parallel trends

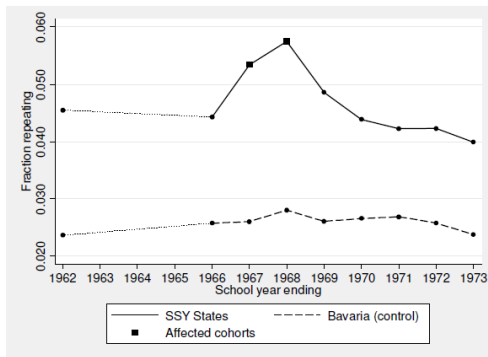
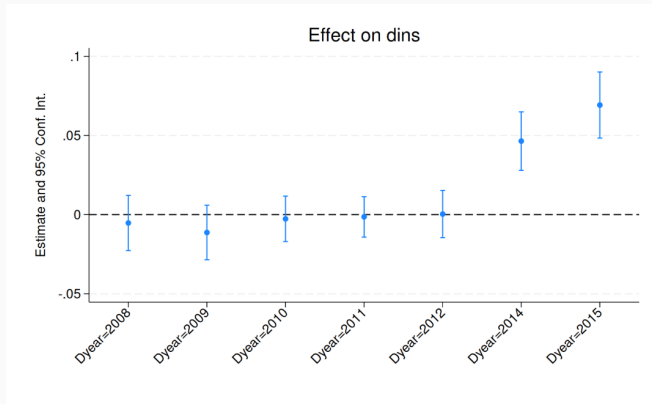


Figure 5.2.3: Average rates of grade repetition in second grade for treatment and control schools in Germany (from Pischke 2007). The data span a period before and after a change in term length for students outside of Bavaria.

Source: Mostly Harmless Econometrics

Common ways to assess parallel trends



Source: Honest DiD - Vignettes

Common ways to assess parallel trends

Placebo tests is essentially a *fake treatment*

- Pick either: a period **before the actual treatment**; a variable **different from the treatment**; an **outcome that shouldn't be affected** by the treatment

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Common ways to assess parallel trends

Placebo tests is essentially a *fake treatment*

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- If you find a non-significant (near zero) estimate: it **supports the credibility** of your estimations

Common ways to assess parallel trends

Placebo tests is essentially a *fake treatment*

- Pick either: a period **before the actual treatment**; a variable **different from the treatment**; an **outcome that shouldn't be affected** by the treatment
- Run the DiD as if it was the real treatment
- If you find a non-significant (near zero) estimate: it **supports the credibility** of your estimations
- If you find a significant effect: it suggests that your DiD may be picking up **pre-existing trends** rather than the effect of your treatment

What if I have pre-trends?

Table 1

A checklist for DiD practitioners.

- Is everyone treated at the same time?

If yes, and panel is balanced, estimation with TWFE specifications such as (5) or (7) yield easily interpretable estimates.

If no, consider using a “heterogeneity-robust” estimator for staggered treatment timing as described in Section 3. The appropriate estimator will depend on whether treatment turns on/off and which parallel trends assumption you’re willing to impose. Use TWFE only if you’re willing to restrict treatment effect heterogeneity.

- Are you sure about the validity of the parallel trends assumption?

If yes, explain why, including a justification for your choice of functional form. If the justification is (quasi-)random treatment timing, consider using a more efficient estimator as discussed in Section 6.

If no, consider the following steps:

1. If parallel trends would be more plausible conditional on covariates, consider a method that conditions on covariates, as described in Section 4.2.
2. Assess the plausibility of the parallel trends assumption by constructing an event-study plot. If there is a common treatment date and you’re using an unconditional parallel trends assumption, plot the coefficients from a specification like (16). If not, then see Section 4.3 for recommendations on event-plot construction.
3. Accompany the event-study plot with diagnostics of the power of the pre-test against relevant alternatives and/or non-inferiority tests, as described in Section 4.4.1.
4. Report formal sensitivity analyses that describe the robustness of the conclusions to potential violations of parallel trends, as described in Section 4.5.

- Do you have a large number of treated and untreated clusters sampled from a super-population?

If yes, then use cluster-robust methods at the cluster level. A good rule of thumb is to cluster at the level at which treatment is independently assigned (e.g. at the state level when policy is determined at the state level); see Section 5.2.

If you have a small number of treated clusters, consider using one of the alternative inference methods described in Section 5.1.

If you can’t imagine the super-population, consider a design-based justification for inference instead, as discussed in Section 5.2.

Source: Roth et al. (2023)

Instructions for tomorrow

- Create an account in UK Data Service: **Instructions**
- How to download data: **Tutorial**
- Study number: 8715
- Download the Stata version

UK longitudinal data

Aim of this session

Working as a Research Fellow I had the experience to **work with different UK datasets** (e.g. Understanding Society, UK Labour Force Survey, Next Steps, British Social Attitudes Survey)

Share my experience and a few tips → all that I would have liked to know before I started

Hopefully to help you **save some time and hassle**

- Very good survey data available on the UK
- Examples of main longitudinal data → BHPS and Understanding Society, UK LFS¹
- Longitudinal cohort studies → e.g. Next Steps, Millennium Cohort Study
- How to access them: the [UK Data Service](#) is one of the main data repository and a great source of information and training!

¹Although this is a rotating panel

UK Data Service - find your data

Home › Find data

Find data

Catalogue search guidance

Our video guide provides step-by-step instructions for using the data catalogue.

[Short tutorial on how to access and download data >](#)

Browse and access key data

Find and access key data easily by themes, data types and teaching datasets.

[Browse and access data >](#)

Access requirements and processes

Open, safeguarded or controlled data access requirements and application guidance for UKDS SecureLab.

[Open, safeguarded or controlled SecureLab access >](#)

[Search](#)

Nesstar explore variables

UK Data Service

Delivering quality social and economic data resources

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Research Datasets

- 1970 British Cohort Study
- Attitudes towards Devolution
- British Election Study
- British Social Attitudes Survey
 - British Social Attitudes Survey, 2022
 - British Social Attitudes Survey, 2021
 - British Social Attitudes Survey, 2020
 - British Social Attitudes Survey, 2019
 - British Social Attitudes Survey: Emergency Care Module, 2018
 - British Social Attitudes Survey, 2018
 - British Social Attitudes Survey, 2017
 - British Social Attitudes Survey, 2016
- Metadata
 - Variable Description
 - Introduction
 - Household Grid
 - Newspaper Readership
 - Party Identification
 - Public Spending and Social Security
 - Health
 - Housing
 - Education
 - Transport
 - Trust in Statistics
 - General Issues
 - Transgender
 - Political and Social Attitudes
 - Employment
 - Trade Unions
 - Retirement and Pensions
 - Classification
 - End Bit
 - Self-Completion
 - Administrative Variables
 - Weighting Variables
 - User Defined variables
- British Social Attitudes Survey, 2015
- British Social Attitudes Survey, 2014
- British Social Attitudes Survey, 2013
- British Social Attitudes Survey, 2012
- British Social Attitudes Survey, 2011
- British Social Attitudes Survey, 2010
- British Social Attitudes Survey, 2009
- British Social Attitudes Survey, 2008
- British Social Attitudes Survey, 2007
- British Social Attitudes Survey, 2006
- British Social Attitudes Survey, 2005
- British Social Attitudes Survey, 2004
- British Social Attitudes Survey, 2003
- British Social Attitudes Survey, 2002
- British Social Attitudes Survey, 2001

DESCRIPTION | **TABULATION** | ANALYSIS

Feedback helps us improve our service to you. Answer two questions about the UK Data Service for the chance to win £50.

[YES](#) [NO THANKS](#)

Dataset: British Social Attitudes Survey, 2016

- Introduction
- Household Grid
- Newspaper Readership
- Party Identification
- Public Spending and Social Security
- Health
- Housing
- Education
- Transport
- Trust in Statistics
- General Issues
- Transgender
- Political and Social Attitudes
- Employment
- Trade Unions
- Retirement and Pensions
- Classification
- End Bit
- Self-Completion
- Administrative Variables
- Weighting Variables

Nesstar - explore sample

Not secure nesstar.ukdataservice.ac.uk/webview/login=

UK Data Service
Delivering quality social and economic data resources

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DESCRIPTION | TABULATION | ANALYSIS

Feedback helps us improve our service to you. Answer two questions about the UK Data Service for the chance to win £50.

YES NO THANKS

Dataset: British Social Attitudes Survey, 2016

Variable RaceOri3: To which of these racial groups do you consider you belong?

LITERAL QUESTION
To which of these groups do you consider you belong? BLACK: of African origin BLACK: of Caribbean origin BLACK: of other origin (WRITE IN) ASIAN: of Indian origin ASIAN: of Pakistani origin ASIAN: of Bangladeshi origin ASIAN: of Chinese origin ASIAN: of other origin (WRITE IN) WHITE: of any origin MIXED ORIGIN (WRITE IN) OTHER (WRITE IN)

Values	Categories	N	%
1	BLACK: of African origin	41	1.4%
2	BLACK: of Caribbean origin	23	0.8%
3	BLACK: of other origin (WRITE IN)	1	0.0%
4	ASIAN: of Indian origin	51	1.7%
5	ASIAN: of Pakistani origin	23	0.8%
6	ASIAN: of Bangladeshi origin	5	0.2%
7	ASIAN: of Chinese origin	18	0.6%
8	ASIAN: of other origin (WRITE IN)	27	0.9%
9	WHITE: of any origin	2689	91.4%
10	MIXED ORIGIN (WRITE IN)	32	1.1%
11	OTHER (WRITE IN)	26	0.9%
98	Don't know	2	0.1%
99	Refusal	4	0.1%
-2	Schedule not applicable	0	
-1	Item not applicable	0	

SUMMARY STATISTICS
Valid cases 2942
Missing cases 0
This variable is numeric

INTERVIEWER INSTRUCTIONS
CARD R7

UNIVERSE
ASK ALL:

- Respondent NS-SEC Socio-economic Class (analytic class level) :dv
- dv: SOC 2010 Sub-Major Group - Respondent
- Respondent SOC2000 sub major group :dv
- Spouse/partner economic activity DV
- Current economic position of spouse/partner
- Spouse : social class[pre-SOC2000]best estimate dv
- Spouse occupational class [7 cat]dv
- NS-SEC analytic classes [spouse] dv
- Spouse NS-SEC Socio-economic Class (analytic class level) :dv
- Spouse Employment status SOCCODE.E52010
- Spouse Employment status SOCCODE.E52000
- Spouse:SEG[pre-SOC2000]best est
- SOC 2010 Sub-Major Group - Partner:dv
- Trade Unions
- Retirement and Pensions
- Classification
- National Identity
- Where were you born?
- Do you see yourself as more English/Scottish/Welsh than British?
- Do you think of self more English or British (England only) DV
- Religion
- National Identity 2
- Does R think of self as British?
- Does R think of self as English?
- Does R think of self as European?
- Does R think of self as Irish?
- Does R think of self Northern Irish?
- Does R think of self as Scottish?
- Does R think of own nationality as Ulster?
- Does R think of self as Welsh?
- Does R think of self as Asian?
- Does R think of self as African?
- R think of self as OTHER nationality?
- R think self none of these nationalities?
- Which one best describes the way you think of yourself?
- Nationality that best describes respondent? DV
- To which of these racial groups do you consider you belong?
- Health and Disability
- Responsibility for Children in the Past
- Education
- EU Referendum
- Income and Benefits

UK Data Service - training

← ↻ 🔍 <https://ukdataservice.ac.uk/training/index> 🌐 ⭐ 🌐 ⋮ 🌐

Enhance your data skills and teaching

[New to using data](#)

Best practice and training for researchers new to accessing and using data in our collection. Includes advice and tools to correctly cite data; student-specific information on our Dissertation Award for undergraduates; and more.

[Data skills modules](#)

There is a wealth of data available for reuse in research and reports. These free, interactive tutorials are designed for anyone who wants to start using secondary data. They show you how to get started with finding good quality data, understanding it and starting your analyses.

[Students](#)

Students can access most of the UK Data Service's collection of social, economic and population data. Find resources to help you find and use our data during your studies including the UK Data Service dissertation resources.

[Survey data](#)

Survey data, including data from long-running surveys, series and longitudinal studies, are a major part of social science research. Learn how to use survey and longitudinal data through training resources including videos, on-demand webinars and written guides.

[International data](#)

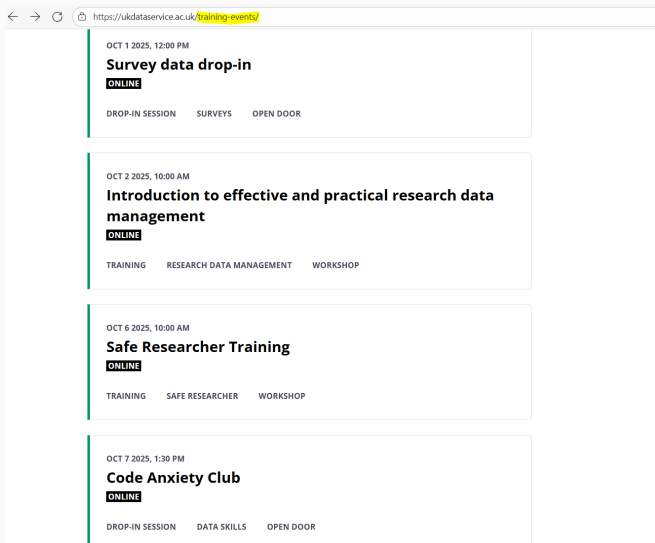
Our international macrodata contain socio-economic time series data aggregated to a country or regional level for a range of countries over a substantial time period.

[Qualitative data](#)

Qualitative research gives a voice to the lived experience, offering researchers a deeper insight into a topic or individuals' experiences. Qualitative data can be combined with quantitative to enhance understanding around a policy or topic in a way that quantitative data by itself often cannot.



UK Data Service - training



A screenshot of a web browser displaying the UK Data Service training events page. The browser's address bar shows the URL <https://ukdataservice.ac.uk/training-events/>. The page lists four training events, each in a white box with a green vertical bar on the left. Each event includes a date and time, a title, a status (ONLINE), and a list of topics.

- OCT 1 2025, 12:00 PM**
Survey data drop-in
ONLINE
DROP-IN SESSION SURVEYS OPEN DOOR
- OCT 2 2025, 10:00 AM**
Introduction to effective and practical research data management
ONLINE
TRAINING RESEARCH DATA MANAGEMENT WORKSHOP
- OCT 6 2025, 10:00 AM**
Safe Researcher Training
ONLINE
TRAINING SAFE RESEARCHER WORKSHOP
- OCT 7 2025, 1:30 PM**
Code Anxiety Club
ONLINE
DROP-IN SESSION DATA SKILLS OPEN DOOR

About the safe researcher training...

Not all data are free to access, and in certain circumstances you will need to file an application and/or be an accredited researcher. The **level of access** depends on the sensitivity/confidentiality of the data.

- Open data → no registration needed
- Safeguarded data → only need to sign the End Users Licence Agreement
- Controlled data: within these we have
 - Secure Licence Data → need to file an application and sign the Secure Licence Agreement
 - Secure Access data → need to file an application and to complete the Safe Researcher Training (SRT).

- Applications for Secure Access data **usually take time** → make sure to factor it in your research plan
- **All members** of your research team have to **complete the SRT** (even if they do not plan to access the data)
- Once your application is successful, you'll **only be able to access the data using the Safe Research Environment**.

Depending on the data you need, you'll either:

- Access the data using a University device which has been checked and approved during the application (needs a static VPN)
- Access the data using a Safe Pod/Safe Point

In any case **the output of the analysis must be cleared before it can be shared**. This also applies to your research team, i.e. if your research team does not access the data, you can't share the analysis output with them before clearance.

Some tips on the application process

The application process is generally **time consuming**. Some advice that could make it a bit smoother:

- **Reply to all application's questions** and items and make sure to download the latest application version
- **Be very specific** in your application (e.g. state your research questions and hypothesis very clearly)
- **Specify the study number and name** rather than just the study name (e.g. SN 8656 Next Steps: Sweeps 1-9, 2004-2023: Secure Access rather than Next Steps Secure Access)
- **Apply for ethical approval** at your University beforehand
- Make the most of your **Safe Research Training**
- **Be prepared to revise it**

Some general tips

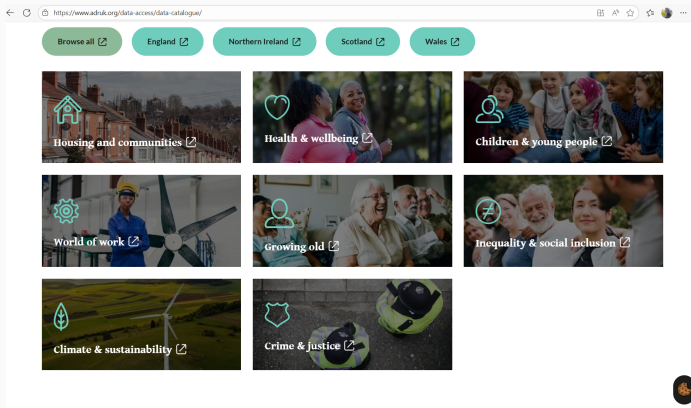
- Make the most of your **Safe Research Training** also in **preparing the output to be cleared** if working with Secure data
- When working with survey data, make sure to use the **appropriate weights** in your analysis
- **Data specific websites** are a very useful source of information (e.g. Understanding Society website or UCL Center for Longitudinal Studies)
- Make sure you are **familiar with the survey guideline and documentation**

Administrative data is information created when people interact with public services, such as schools, the NHS, the courts or the benefits system, and collated by government.

These public bodies must keep records of these interactions for operational purposes: to enable them to carry out their day-to-day work, to monitor and improve their performance, and keep providing services in an effective way.

- Important source of information for social sciences
- Some examples: Longitudinal Education Outcomes (LEO data), Annual Survey of Hours and Earnings (ASHE) linked to HMRC and 2011 Census
- Not self-reported → less measurement error
- Bigger numbers → more precise estimations and more possibility to conduct sub-group analysis
- More confidentiality issues → controlled data

Administrative Data Research UK

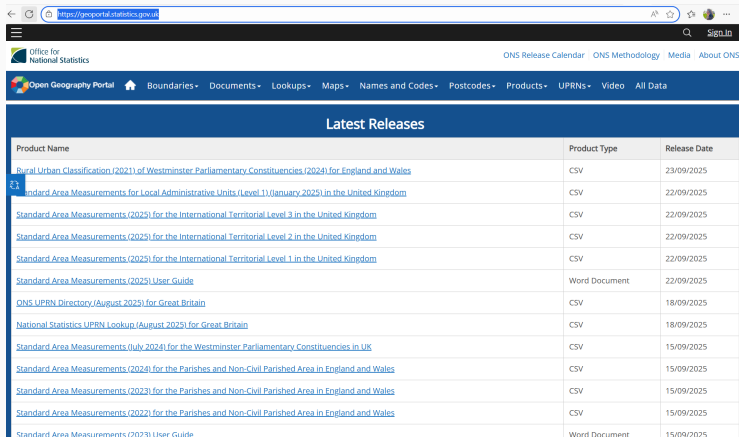


More on administrative data

- Mainly available through Safe Pods or trusted research environment
- Data provider is the ONS
- Application process similar to the survey data Secure Access, slightly more complicated → more time consuming
- Apparently quite complicated to process and analyse
- Only available to Accredited Researchers

Geography data

- Often survey and/or administrative data will allow us to carry out analysis at a lower geographical level (e.g. regions or Local Authority Districts)
- Main source:



The screenshot shows the Open Geography Portal website. The header includes the Office for National Statistics logo and navigation links like 'ONS Release Calendar', 'ONS Methodology', 'Media', and 'About ONS'. The main navigation bar lists categories such as 'Boundaries', 'Documents', 'Lookups', 'Maps', 'Names and Codes', 'Postcodes', 'Products', 'UPRNs', 'Video', and 'All Data'. The 'Latest Releases' section contains a table of recent data products.

Product Name	Product Type	Release Date
Rural Urban Classification (2021) of Westminster Parliamentary Constituencies (2024) for England and Wales	CSV	23/09/2025
Standard Area Measurements for Local Administrative Units (Level 1) (January 2025) in the United Kingdom	CSV	22/09/2025
Standard Area Measurements (2025) for the International Territorial Level 3 in the United Kingdom	CSV	22/09/2025
Standard Area Measurements (2025) for the International Territorial Level 2 in the United Kingdom	CSV	22/09/2025
Standard Area Measurements (2025) for the International Territorial Level 1 in the United Kingdom	CSV	22/09/2025
Standard Area Measurements (2025) User Guide	Word Document	22/09/2025
ONS UPRN Directory (August 2025) for Great Britain	CSV	18/09/2025
National Statistics UPRN Lookup (August 2025) for Great Britain	CSV	18/09/2025
Standard Area Measurements (July 2024) for the Westminster Parliamentary Constituencies in UK	CSV	15/09/2025
Standard Area Measurements (2024) for the Parishes and Non-Civil Parished Area in England and Wales	CSV	15/09/2025
Standard Area Measurements (2023) for the Parishes and Non-Civil Parished Area in England and Wales	CSV	15/09/2025
Standard Area Measurements (2022) for the Parishes and Non-Civil Parished Area in England and Wales	CSV	15/09/2025
Standard Area Measurements (2023) User Guide	Word Document	15/09/2025

Stat-Xplore Home

Search

Select dataset or table

Datasets

- Stat-Xplore
 - Alternative Claimant Count
 - Alternative Claimant Count
 - Alternative Claimant Count Off Flows
 - Alternative Claimant Count On Flows
 - Attendance Allowance
 - AA: Cases in Payment - Data from May 2
 - AA: Cases in Payment - Data to February
 - AA: Cases with entitlement - Data from 1/
 - AA: Cases with entitlement - Data to Feb
 - Benefit Cap
 - HB Cumulative Caseload
 - HB Cumulative Caseload - 2001 COA (u)
 - HB Point in Time Caseload
 - HB Point in Time Caseload - 2001 COA (l)
 - UC Cumulative Caseload
 - UC Point in Time Caseload
 - Benefit Combinations
 - Benefit Combinations - Data from May 20
 - Benefit Combinations - Data from May 20
 - Benefit Combinations - Data to February
 - Bereavement Benefits
 - Bereavement Benefits - Data from May 2
 - Bereavement Benefits - Data to February
 - Bereavement Support Payment
 - BSP: Claims in Payment
 - BSP: Claims Received
 - Carers Allowance
 - CA: Cases in Payment - Data from May 2
 - CA: Cases in Payment - Data to February
 - CA: Cases with entitlement - Data from 1/
 - CA: Cases with entitlement - Data to Feb
 - Child Maintenance Service

Tables

Select a dataset to see available tables

Welcome to Stat-Xplore

Stat-Xplore provides a guided way to explore Department for Work and Pensions (DWP) benefit statistics, currently holding data on benefits as seen in the interactive dataset list on the left-hand panel. In the future, Stat-Xplore will include data on a wider set of DWP benefits and programmes.

Getting Started

Use Stat-Xplore to create your own custom tabulations and view the results in interactive charts. Simply click once on the data set you would like to view on the left-hand side to interact with ready-made tables. Alternatively, double click on the dataset and you will be able to build your bespoke table.

User Guidance

User guidance is available in the ["help"](#) section which can be found in the top right hand side with a question mark icon or at the bottom right hand side with the label "Need Help?".

The following [video](#) outlines some of the basic functionality available within Stat-Xplore online tool and shows users how to create and edit a simple table, as well as other basic features available within the tool such as downloading data in different formats, saving a user-made table and also how to present data using different chart options. Note that not all the functionality is available unless you are a registered user, such as saving tables.

Data Confidentiality

For the datasets available on Stat-Xplore, statistical disclosure control is applied which guards against the identification of an individual claimant. For further information see [Data Confidentiality](#).

The introduced random error method applied to the data from August 2024 will have a small impact on already published historic figures across Stat-Xplore releases. This update was necessary to enable a move to a more advanced data processing platform to maintain data confidentiality.

Moving to Census Output Area 2021 Geographies

Unless specified, most of the geographies available on the Stat-Xplore datasets currently use the 2011 Census Output Areas (COAs). Development work and testing is underway to move our statistical publications to report on 2021 COA geographies. This will be on a rolling basis starting for some publications from Autumn 2025. Users will be informed of updates via the [DWP Statistical Work Programme](#) and updates to this page. We will also take the opportunity to standardise the range of COA 21 affected geographies.

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Welcome to Nomis

Nomis is a service provided by [Office for National Statistics \(ONS\)](#), the UK's largest independent producer of official statistics. On this website, we publish statistics related to population, society and the labour market at national, regional and local levels. These include data from current and previous censuses.

[Information for first-time visitors](#)
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Labour Market Profiles

View a labour market profile of an area. Includes some of the data from our key datasets on population, employment, unemployment, qualifications, earnings, benefit claimants and businesses.

- [Local Authority Profile](#) (district/county areas)
- [Local Enterprise Partnerships Profile](#)
- [Combined Authority Profile](#)
- [Regional and National Profile](#)
- [2010 Parliamentary Constituencies Profile](#)
- [2011 Ward Profile](#) (England & Wales only)

2021 Census Local Area Report

View a [2021 Census report for a local area](#) such as a ward, village or town. Includes information on the characteristics of people and households in the area.

Data Downloads

Create a data download from one of our full range of data sets. Data is available at a very detailed level.

[Query data](#)
 Download figures from a single data set.

Census Statistics

[2021 Data catalogue](#)
 Browse by table type and number.

[2021 Search by topic \(table finder\)](#)
 Search by keyword and geography type.

[2011 Data catalogue](#)
 Browse by table type and number, or view by release.

[2011 Search by topic \(table finder\)](#)

Tips on working with geographical data

- These data are usually **open and under OGL Licence** (mention it in your application if you want to link them to Controlled Access or other data)
- **LADs codes changed between 2011 and 2021** → Lookup table on geoportal
- UK lower level area classification is **slightly different from the one used in Europe** (i.e. no direct correspondence with NUTS level geography)

Thank you for your
attention!

Feel free to get in touch:
valentina.diasio@soton.ac.uk